

Finn Hertsch

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Research interests: representation learning & retrieval, numerically robust online learning, computer vision for planetary science

EDUCATION

B.Sc. Data Science and Artificial Intelligence (210 ECTS)

10/2024 – 2026 (expected)

DHBW Ravensburg Ravensburg, Germany

- GPA: 2.3 | Cooperative (dual) program combining coursework with applied industry practice
- Focus areas: advanced programming & systems engineering, machine learning, MLOps
- Capstone project: hybrid article recommendation system for a regional media group, deployed on Google Cloud Platform (score: 97/100)

Abitur (general university entrance qualification) 2021 – 2024

Edith-Stein-Schule Ravensburg, Germany

- Focus subject: biotechnology | Final grade: 2.0
- Bioinformatics coursework: sequence alignment and database search using BLAST and the FASTA format/algorithm
- Molecular biology practicals: GFP reporter-gene transformation in *E. coli*
- Bioprocess engineering practicals: fermentative citric acid biosynthesis in a bench-scale bioreactor

RESEARCH EXPERIENCE

Luna — Planetary-Scale Lunar Pit Detection 2026 – ongoing

Independent research collaboration with Prof. Dr.-Ing. Mark Schutera, DHBW Ravensburg Ravensburg, Germany

- Designed and trained HOLE, a DINOv3 ViT-S/16 backbone with a LoRA adapter and a custom Matryoshka-DIVE loss for lunar surface representation learning
- Built a coverage-optimization solver selecting a minimal set of LROC NAC images for $\geq 99\%$ global lunar surface coverage across 7 latitude bands, processing over 20,000 images end-to-end
- Designed and implemented Pithos, a GraalVM-native, CUDA-accelerated vector database serving as the retrieval backend, indexing over 125 million vectors at ~ 5.1 MB/product
- Set up planetary image-processing infrastructure (ISIS3) and a full annotation/dataset pipeline for lunar surface segmentation, maintained as a public Hugging Face dataset

Kairos — Numerically Robust News Recommendation under Item Cold-Start 2026

DHBW Ravensburg Ravensburg, Germany

- Proposed a Cholesky-factor-based LinUCB update as a numerically stable alternative to Sherman–Morrison inversion for contextual bandits under sparse, high-churn data
- Integrated Matryoshka Representation Learning for adaptive-latency candidate retrieval, achieving a $4.78\times$ inference speedup at $< 3.2\%$ approximation error
- Validated the framework on live data from the Tagesschau API

PUBLICATIONS

- F. Hertsch, “Kairos: Numerisch robuste News Recommendation unter Item-Cold-Start mit Cholesky-basiertem LinUCB.” Submitted to SKILL 2026 (under review).

PROFESSIONAL EXPERIENCE

Founder & Managing Directorsince 02/2026

Hertsch Technologies UG (haftungsbeschränkt)Germany

- Founded and lead a company focused on automated data-processing pipelines and professional maintenance infrastructure

Dual-Study Practical Partner2026 – 09/2027

Waldner HoldingWangen, Germany

- Applied data science and software engineering work as part of the DHBW cooperative study program

Dual-Study Practical Partner10/2024 – 06/2026

Schwäbischer Verlag GmbH & Co. KGRavensburg, Germany

- Developed a hybrid recommendation system (NCF & CBF) on Google Cloud Platform, deployed on Vertex AI
- Built data pipelines processing user interaction data with BigQuery
- Prototyped a multi-armed contextual bandit over high-dimensional article embeddings

TECHNICAL SKILLS

- **Programming:** Python, Java, SQL
- **ML & Data:** PyTorch-based model training (DINOv3, LoRA), MLOps (DASC-PM), hybrid filtering, vector search / ANN
- **Cloud & Infrastructure:** Google Cloud Platform (Vertex AI, BigQuery), GraalVM, CUDA, FastAPI, PowerBI

LANGUAGES

- German (native) | English (B2/C1) | Spanish (B1)